

such that

$$|B(x) - 2B_0^+(x)| \leq C^+ \quad \text{and} \quad |W(x) - 2B_0^-(x)| \leq C^-$$

uniformly for all irrational numbers x .

The theorem can be deduced from the work of Wilton, by a modification of his application of Voronoi's summation formula, but the proof we present here will be elementary in the sense that it relies on no complex analysis, and in fact only on elementary properties of α -continued fractions as established, e.g., in [LMNN10].

Since the odd part B^- and the even part W^+ of the Brjuno respectively Wilton functions are uniformly bounded functions — see Proposition 2.1 for closed form expressions — the functions of interest following Theorem 1.1 are the differences

$$\Delta^+(x) := B^+(x) - 2B_0^+(x) \quad \text{and} \quad \Delta^-(x) := W^-(x) - 2B_0^-(x).$$

Numerical computations show that $\Delta^\pm$ behave widely differently.

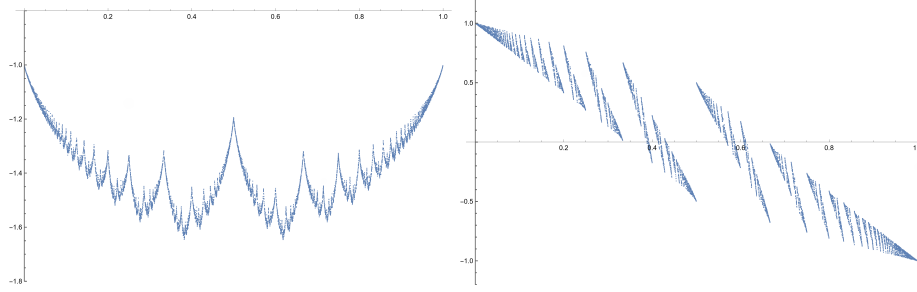


FIGURE 3. Plots of Δ^+ and Δ^- at 10000 randomly chosen points

In [LM21, Theorem 1], the second and third authors established that $\Delta^+(x)$ extends to a $\frac{1}{2}$ -Hölder continuous function on the real line. This is in stark contrast to what one observes in the graphical plots of $\Delta^-(x)$, which seems to indicate that $\Delta^-(x)$ is discontinuous at each rational with a jump of order $2/q$ for $x = p/q$. In that respect, it is reminiscent of the ‘popcorn function’

$$f(x) = \begin{cases} 1/q & \text{if } x = p/q \text{ with } (p, q) = 1; \\ 0 & \text{if } x \text{ is irrational;} \end{cases}$$

that staple of calculus illustrating the existence of a Riemann integrable function with infinitely many discontinuities. We confirm these numerical observations with the following theorem.

Theorem 1.2. *The function $\Delta^-(x)$ is continuous on $(0, \frac{1}{2}) \setminus \mathbb{Q}$. At each rational $x = p/q$, it has an increasing jump of $2/q$. In particular $\Delta^-(x)$ is Riemann integrable and nowhere differentiable.*